

EXPERIMENT - 2

Develop a JavaScript Quiz Application with Timer and Scoring System Using DOM Manipulation

OBJECTIVE OF THE EXPERIMENT:

To develop an interactive quiz application using JavaScript that includes a timer, scoring system, and DOM manipulation for dynamic question display and user interaction.

OUTCOME OF THE EXPERIMENT:

Students will be able to create a dynamic quiz application, manipulate HTML elements using the DOM, implement a countdown timer, and calculate scores using JavaScript.

SYSTEM REQUIREMENTS:

- HTML5-compliant Browser
- Text Editor/IDE (e.g., VS Code)

ALGORITHM / PROCEDURE:

1. Create the HTML structure for the quiz.
2. Style the application using CSS.
3. Define quiz questions and answers in JavaScript.
4. Display questions dynamically using DOM manipulation.
5. Start the timer when the quiz begins.
6. Check the selected answer and update the score.
7. Display the next question until all questions are completed.
8. Show the final score after the quiz ends or when the timer expires.

PSEUDOCODE:

START

Create quiz questions and answers.

Initialize score and timer.

Display the first question.

Start countdown timer.

WHILE questions remain AND timer > 0

 Accept user's answer.

 IF answer is correct

 Increase score.

 END IF

Display next question.
END WHILE

Display final score.

END

Sample Code:

HTML:

```
<!DOCTYPE html>
<html>
<head>
  <title>Quiz App</title>
  <link rel="stylesheet" href="style.css">
</head>
<body>

<div class="quiz-container">
  <h2>JavaScript Quiz</h2>

  <div id="timer">Time: 30</div>

  <h3 id="question"></h3>

  <div id="options"></div>

  <button onclick="nextQuestion()">Next</button>

  <h3 id="result"></h3>
</div>

<script src="script.js"></script>

</body>
</html>
```

CSS

```
body{
  font-family: Arial, sans-serif;
  background:#f4f4f4;
  display:flex;
  justify-content:center;
  align-items:center;
  height:100vh;
}

.quiz-container{
  background:white;
  padding:20px;
  width:350px;
  border-radius:10px;
  box-shadow:0 0 10px gray;
}

button{
```

```

margin-top:15px;
padding:10px 20px;
background:#007bff;
color:white;
border:none;
cursor:pointer;
}

button:hover{
background:#0056b3;
}

```

JavaScript:

```

const quiz = [
{
question:"Which language is used for web page interactivity?",
options:["HTML","CSS","JavaScript","Python"],
answer:"JavaScript"
},
{
question:"Which tag creates a heading?",
options:["<p>","<h1>","<img>","<a>"],
answer:"<h1>"
}
];

let index = 0;
let score = 0;
let time = 30;

const question = document.getElementById("question");
const options = document.getElementById("options");
const result = document.getElementById("result");

function loadQuestion(){
question.innerHTML = quiz[index].question;
options.innerHTML = "";

quiz[index].options.forEach(option=>{
options.innerHTML += `<input type="radio" name="ans" value="${option}"> ${option}<br>`;
});
}

function nextQuestion(){
const selected = document.querySelector('input[name="ans"]:checked');

if(selected && selected.value === quiz[index].answer){
score++;
}

index++;

if(index < quiz.length){
loadQuestion();
}
else{
result.innerHTML = "Your Score: " + score + "/" + quiz.length;
question.innerHTML = "";
options.innerHTML = "";
}
}

```

```

}

loadQuestion();

setInterval(function(){
    time--;
    document.getElementById("timer").innerHTML = "Time: " + time;

    if(time==0){
        result.innerHTML="Time Over! Score: "+score+"/"+quiz.length;
        question.innerHTML="";
        options.innerHTML="";
    }
},1000);

```

RESULTS:

A functional JavaScript quiz application with dynamic question display, countdown timer, and scoring system was developed successfully using DOM manipulation.

PRE LAB QUESTIONS:

1. What is DOM manipulation in JavaScript?
2. What is the purpose of `setInterval()`?
3. How can JavaScript access HTML elements?
4. What is the use of event handling in JavaScript?

SAMPLE VIVA VOCE QUESTIONS:

1. What is the Document Object Model (DOM)?
2. Explain the purpose of `getElementById()`.
3. What is the difference between `setInterval()` and `setTimeout()`?
4. How is the score calculated in the quiz application?
5. How does JavaScript dynamically create HTML elements?
6. What is event handling?
7. Why is a timer used in quiz applications?
8. What is the purpose of radio buttons in forms?
9. Explain the role of arrays in the quiz application.
10. How does JavaScript validate the selected answer?